The algorithm leverages the inductive definition of a graph: while trees can be defined by induction on the number of vertices (we construct larger trees by adding children to a root), given a graph $G=(V,E)$ it can be inductively grown in two different ways.

As we have seen in chapter 14, in fact, it could be:

1. $G'=(V+\{v\}, E)$: we add a new vertex to G.
2. $G'=(V, E+\{(u,v)\}) \mid u, v \in V$: we add a new edge to G.

Listing 15.2 Planarity Testing based on KK's theorem

```
function isPlanar(graph)
  if |graph.vertices| < 5 then return
true                              #1
  if violatesEulerConstraints(graph) then return
false                 #2
  if isK5(graph) or isK3_3(graph) then return false
                   #3
  for v in graph.vertices do
                                      #4
    subG ←
graph.remove(v)                                          #5
    if !isPlanar(subG) then return false
                             #6
  for e in graph.edges
do                                             #7
    subG ←
graph.remove(e)                                          #8
    if !isPlanar(subG) return
false                                      #9
  return
true
#10
```